

SabarishMohan

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[LinkedIn](#) | [GitHub](#)

Education

Sri Sivasubramaniya Nadar College of Engineering

2023 - 2027

B. E in Electronics and Communication Engineering

CGPA: 9.101(as of sem 5)

Relevant Coursework: Digital Systems, Verilog, VLSI design, Computer Architecture

Vidhya Vikashni Matric Higher Secondary School

2021- 2023

Class 12 (TN HSC)

Percentage: 97.5%

Relevant Coursework: Physics, Mathematics, Chemistry

Technologies and Skills

Programming Languages & HDL: C, C++, Python, Verilog HDL

Tools & EDA: Xilinx Vivado, Intel Quartus Prime, EDAplayground, Cadence Virtuoso, eSim

Digital design & Verification: RTL Design,FSM Design, Datapath & Control, Testbench Development, Debugging, Simulation

Hardware Platforms: Nexys A7 FPGA Board, Intel De0 Nano FPGA Board

Experience

eSim FOSSE IITB Summer Fellowship

June 2025 - July 2025

- Designed and verified 10+ digital ICs for the eSim simulation platform.
- Developed subcircuit models directly from datasheet specifications, ensuring accuracy and performance.
- Simulated circuit behavior and validated functionality through custom circuits.

Projects

E-yantra Maze Solver Bot Competition – AIR 5

Aug 2025 – March 2026

- Design and Implementation of the autonomous maze solver bot in the eyantra competition.
- Integration of sensors and development of algorithms to solve the maze using Verilog and Intel Cyclone IV E FPGA board.
- Selected for finale

RISC-V Single Cycle Processor

July 2025- Aug 2025

- Designed a single cycle RISC-V (RV32I) processor with complete data path and control unit in Verilog.
- Implemented instruction fetch, decode, execute, memory, and write-back stages.
- Verified instruction execution through simulation and waveform analysis.

Implementation of UART protocol using FPGA

July 2025

- Designed and implemented a UART communication protocol using Verilog on the Nexys 7 FPGA board for serial data transfer.
- Developed modular TX and RX blocks with start/stop bit framing, and synchronization.
- Verified through RTL simulation and FPGA based hardware testing.

RISC-V Assembly Simulation using Ripes Simulator

June 2025

- Simulated basic RISC-V assembly programs in Ripes to understand instruction execution and datapath behavior.
- Observed register, memory and control signal changes using step by step visual simulation.

CNN based Fall Detection system using verilog

Dec 2024-Mar 2025

- Designed and implemented a CNN-based fall detection accelerator in Verilog.
- Implemented Conv2D, Pooling, and Dense layers using fixed-point arithmetic optimized for FPGA

Achievements and Initiatives

- DVCon India 2025 Design Contest - Stage 3** – Selected to Stage 3 (Team Silicon Sprints – No.8071).
- DVCon India 2025 Fellowship** – Selected and participated the Design verification conference as a fellowship delegate.
- Merit Scholarship Recipient** – Awarded college merit scholarship for achieving Department Rank 3 in the first and second year(till sem5).
- Verilog Competitive Programming Contest** – Won first place in an intra-college competition on Verilog HDL design and debugging.
- Attended Digital India RISV Symposium** – IITM

Volunteering

- VLSI Student Coordinator**, SSN Tech Club June 2025 – Current
- NSS Volunteer**, SSN College of Engineering Aug 2023- Current
- Event Organizer**, INVENTE – SSN National-Level Tech Fest March 2025
- Event Manager**, SSN IEEE ComSoc Student Chapter Aug 2024- April 2025